Harmonic Offsets in Voiced Speech

Rujia Li

Supervisor: Andreas Jakobsson

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1 Introduction

Voiced speech is often described as a quasi-harmonic signal: a fundamental frequency f_0 together with spectral peaks near integer multiples nf_0. In a perfectly harmonic model, the nth harmonic would be located exactly at nf_0. In real voice recordings, the peaks are typically slightly shifted due to non-linear vocal fold vibration, coupling with the vocal tract, and short-term variations in phonation. In this project we focus on these small deviations, which we refer to as *harmonic offsets* (or *inharmonicity offsets*).

For each voiced frame, we estimate f_0 and the peak frequency f_n of the nth harmonic, and define

$$\Delta_n = f_n - nf_0, \qquad n = 1, \ldots, 6. \tag{1}$$

We collect the first six offsets into a vector

$$\mathbf{\Delta} = [\Delta_1, \Delta_2, \ldots, \Delta_6]^\top \in \mathbb{R}^6, \tag{2}$$

which summarizes how the observed harmonic structure deviates from an ideal harmonic spectrum within that frame.

This report addresses two related questions. First, *to what extent can harmonic offsets be explained by standard acoustic measures without relying on speaker identity?* Concretely, we test whether there exists a speaker-independent structure between the offset vector $\mathbf{\Delta}$ (and, in practice, Δ_1 as a representative example) and common voice measures such as f_0, harmonic-to-noise ratio (HNR), intensity, and spectral descriptors (e.g., spectral tilt). Answering this question is useful because it may reveal general and interpretable mechanisms behind offset variability, rather than patterns that only reflect idiosyncrasies of particular individuals.

Second, *do different speakers have different offset distributions?* Even if part of $\mathbf{\Delta}$ can be attributed to measurable acoustic covariates, there may still be systematic between-speaker differences in the remaining variability. We therefore test whether speakers differ in the distribution of $\mathbf{\Delta}$, and whether such differences are expressed mainly as mean shifts or, more strongly, as changes in dispersion and covariance structure across the six dimensions.

To study these questions, we build a frame-level pipeline that extracts voiced, vowel-like frames from continuous speech, estimates f_0, and measures the first six harmonic offsets per frame. We then conduct two types of analyses: (i) speaker-independent modeling of offsets using acoustic features with evaluation across speakers, and (ii) speaker-level statistical tests comparing offset distributions across speakers. In summary, we find that acoustic features show consistent but limited explanatory power for offsets at the frame level, while between-speaker differences in offset distributions are strong and appear more robust in variability and covariance patterns than in simple mean offsets.

Throughout the report we emphasize robustness: speaker identity should not leak into "speaker-independent" claims. We therefore evaluate feature–offset associations using out-of-speaker testing, and we study between-speaker differences using multivariate distributional tests that capture dispersion and covariance, not only mean shifts. This combination gives a clearer picture of what can be explained by common acoustic factors versus what remains genuinely speaker-specific.

2 Voice Data and Preprocessing

2.1 Voice data

This project uses ordinary (non-clinical) voice recordings. The dataset contains multiple recordings per speaker, including continuous spoken sentences as well as a smaller set of sustained-vowel style recordings. The number of recordings (and therefore the number of usable voiced frames) varies a lot across speakers, so the data are naturally unbalanced. This is important later when we compare speakers, since we must avoid letting a small number of speakers dominate the statistics.

2.2 Vowel frame extraction and storage

To study harmonic offsets reliably, we want frames that are clearly voiced and vowel-like. Starting from each recording, I first locate candidate vowel regions using a standard alignment-style approach: the audio is segmented using word-level timing information, and vowel-like portions are kept as candidates. From these regions, I cut the signal into short fixed-length frames (typically 20 ms), keeping the original time stamps (start/stop time in seconds) for each frame.

For practical reasons, each extracted frame is saved as an individual `.wav` file. This makes the later feature computation modular and easy to audit: any frame can be listened to, re-processed, or filtered out without rerunning the full segmentation step. The file name encodes the speaker ID, recording type, recording timestamp, and the frame time interval (e.g., `3.090--3.110` seconds).

2.3 Frame-level feature pipeline

After frame extraction, all frames are processed by a single pipeline (implemented in `run_vowel_frame_feature_pipeline`). The goal is to estimate the fundamental frequency f_0 robustly, measure the first six harmonic offsets, and compute a small set of additional acoustic descriptors for later analysis. The main steps are:

1. **Basic quality measures (time-domain).** For each frame, we compute simple indicators such as RMS level, zero-crossing rate, an autocorrelation-based periodicity peak, and an HNR-like measure. Frames that are very weak or weakly periodic are *tagged* (not necessarily removed immediately), so we can later study how filtering affects results.

2. **Adaptive f_0 range.** A rough f_0 estimate from autocorrelation is used to set a tighter, frame-specific search range. This reduces common octave errors in low-pitched or noisy frames.

3. **Robust f_0 estimation with octave-error control.** Estimating f_0 from short speech frames is often affected by *octave errors* (pitch doubling/halving), where the estimator locks onto $2f_0$ or $f_0/2$ instead of the true fundamental. This can happen in speech when the first harmonic is weak (or effectively "missing") while higher harmonics are strong, and when short windows limit frequency resolution and increase sensitivity to noise and spectral leakage. To make f_0 more stable, we apply several safeguards:

   (a) **Quality tagging.** We use simple energy and periodicity indicators (RMS level and an autocorrelation peak) to tag weak or low-periodicity frames, since these are most prone to octave mistakes.

   (b) **Range tightening from autocorrelation.** A rough autocorrelation-based f_0 estimate is used to define a frame-specific search interval, reducing spurious solutions far from the plausible pitch range.

   (c) **Two-stage OMT-regularized harmonic fit.** We fit a harmonic model with an optimal-mass-transport (OMT)–inspired regularizer, which penalizes deviations from an ideal harmonic grid while still allowing small, data-supported inharmonicity. This formulation follows the general idea introduced in [2, 1]. In practice, we run the fit in two stages: a first pass with a stronger prior to stabilize the f_0 estimate, followed by a second pass with a looser regularization for refinement.

   (d) **Explicit octave-candidate check.** We test a small candidate pool containing f_0, $0.5f_0$, and $2f_0$ (and the same variations around the autocorrelation mode), score candidates by a harmonic-sum criterion over multiple harmonics, and then refine the selected candidate in the final pass.

4. **Harmonic peak refinement and offsets.** Given the final f_0, we estimate the *observed* harmonic peak frequency f_n by searching the magnitude spectrum in a narrow band around the expected location nf_0. Within each band, we take the strongest local spectral peak and refine its frequency using a simple log-magnitude parabolic interpolation (sub-bin refinement). The harmonic offset is then measured in Hz as

$$\Delta_n = f_n - nf_0, \qquad n = 1, \ldots, 6.$$

(Frames/harmonics for which no clear peak is found in the search band are marked as missing and excluded from later analyses that require complete offsets.)

5. **Additional spectral features.** We compute a compact set of descriptors that are commonly used as proxies for phonation and spectral shape, such as spectral centroid/spread/tilt and harmonic amplitude ratios (e.g., $H1 - H2$, $H2 - H4$).

2.4 Output table format

The pipeline outputs one row per frame in a CSV file. Each row contains metadata (speaker, recording ID, time stamps), f_0 diagnostics, the six offsets, additional acoustic features, and quality flags. Table 1 shows a shortened example of the table schema (the full table contains more columns of the same kind).

Table 1: Example columns in the frame-level feature table (shortened).

wavName	**speakerID**	**recType**	**timestamp**	**startSec**	**stopSec**	**durationSec**
`A_..._3.090-3.110_20ms.wav`	`Z000XZYR`	`t`	`220831...`	3.090	3.110	0.020

f_0 **(final)**	$\mathbf{\Delta_1}$	$\mathbf{\Delta_2}$	$\cdots$	$\mathbf{\Delta_6}$	**hnr_dB**	**H1H2_dB**	**isWeak**
121.4	-1.2	3.5	$\cdots$	-0.8	18.6	-2.1	0

In addition to these example columns, the table includes intermediate f_0 estimates (for debugging), spectral descriptors (centroid, spread, tilt, high-band energy ratio), and flags that mark octave errors and missing/invalid offset estimates.

2.5 Basic data summary

After preprocessing, the analysis dataset consists of a large collection of short voiced frames, each represented by a 6-dimensional offset vector $\mathbf{\Delta}$ together with a small set of acoustic features and quality flags. There are two noteworthy aspects to the data. First, the number of extracted frames per recording and per speaker is highly variable, which motivates later balancing steps (e.g., speaker filtering and downsampling) when performing speaker-level tests. Second, although the offsets are computed frame-by-frame, the downstream questions in this report are mainly distributional: we study how offsets relate to acoustic features in a speaker-independent way, and we test whether speakers exhibit different offset distributions, especially in variability and covariance structure.

3 Speaker-independent feature–offset relationships

This section addresses the first research question: *to what extent can harmonic offsets be explained by standard acoustic measures without relying on speaker identity?* All analyses are performed at the frame level using short voiced frames of length 20 ms.

3.1 Target variable and predictors

In principle, each frame is represented by a 6-dimensional offset vector $\mathbf{\Delta} = [\Delta_1, \ldots, \Delta_6]^\top$, where Δ_k is the deviation (in Hz) of the k-th harmonic peak from the ideal location kf_0. To keep the analysis focused and interpretable, we use the **first-harmonic offset** Δ_1 as a representative example. Empirically, Δ_1 is

the most stable component: it is less affected by weak high-order harmonics, local spectral notches, and peak-picking failures, and therefore tends to have fewer missing or extreme values than higher-order offsets.

As predictors, we use a compact set of standard voice measures and spectral descriptors available in the frame-level table:

- fundamental frequency estimate: `f0_final`,
- harmonic-to-noise ratio proxy: `hnr_dB`,
- intensity proxy: `rms_dB`,
- harmonic amplitude ratios: `H1H2_dB`, `H2H4_dB`,
- spectral shape: `specCentroid`, `specSpread`, `specTilt`,
- high-band energy ratio: `hiBand14k_ratio`.

Before modeling, predictors are standardized (z-scored) to make coefficients comparable and to improve numerical stability. Frames with missing values in the target or selected predictors are removed for the corresponding model fits.

3.2 Speaker-independent evaluation protocol

A central point in this project is to test *true* speaker-independent (SI) generalization. If frames are randomly split into training and test sets, the same speaker will typically appear in both sets. In that case, the model can implicitly exploit speaker-specific baselines (and other idiosyncrasies), leading to overly optimistic performance estimates that do not answer the SI question.

We therefore adopt **speaker-grouped K-fold cross-validation** using `speakerID` as the grouping variable (GroupKFold). Concretely, we partition the set of speakers into $K = 5$ folds. For each fold $k \in \{1, \ldots, K\}$, we train the model on all frames from speakers in the other $K - 1$ folds and evaluate on *all* frames from the held-out speakers in fold k. This ensures *no speaker overlap* between training and test:

$$\mathcal{S}_{\text{train}}^{(k)} \cap \mathcal{S}_{\text{test}}^{(k)} = \varnothing,$$

where $\mathcal{S}_{\text{train}}^{(k)}$ and $\mathcal{S}_{\text{test}}^{(k)}$ denote the speaker sets used in fold k.

Metrics and baselines. We report performance by averaging metrics across folds. For each fold we compute the coefficient of determination R^2 and mean absolute error (MAE, in Hz) on the held-out speakers, and then report the cross-validated mean. Because Δ_1 exhibits heavy-tailed residuals, MAE is often more stable and interpretable than squared-error measures.

In addition to model baselines (e.g., OLS), we also report a *median baseline* that predicts a constant: the training-fold median of $y = \Delta_1$. This is the optimal constant predictor under absolute loss, and provides a simple "no-feature" reference.

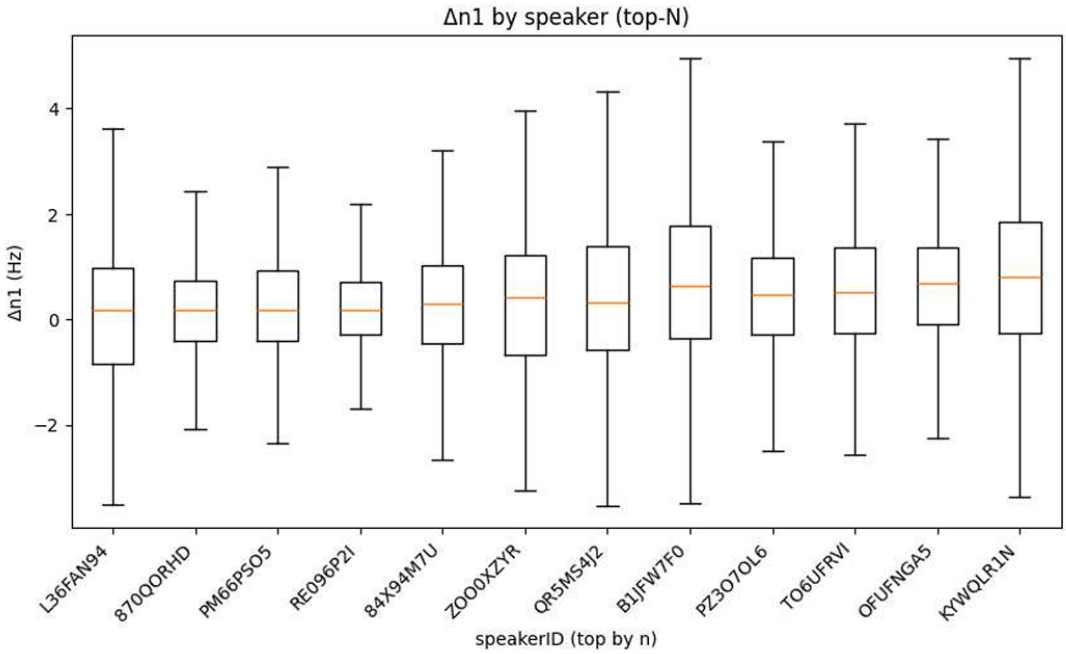


Figure 1: Distribution of Δ_1 across speakers (illustrative). Most speakers are centered near zero, suggesting that mean shifts alone are not strongly discriminative.

3.3 Linear baseline: ordinary least squares (OLS)

We start with multiple linear regression as a simple, interpretable baseline. Let $y_i = \Delta_{1,i}$ denote the first-harmonic offset for frame i, and let $\mathbf{x}_i \in \mathbb{R}^p$ be the standardized predictor vector. The OLS model is

$$y_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \tag{3}$$

where $\boldsymbol{\beta}$ is estimated by minimizing squared error on the training folds.

SI performance. Under speaker-grouped CV, OLS achieves

$$R^2_{\mathrm{SI}} \approx 0.0034, \qquad \mathrm{MAE}_{\mathrm{SI}} \approx 1.239 \text{ Hz},$$

only marginally better than the median baseline ($R^2_{\mathrm{SI}} \approx -0.0127$, $\mathrm{MAE}_{\mathrm{SI}} \approx 1.249\,\mathrm{Hz}$). In other words, **linear explainability is negligible** under SI evaluation.

In-sample fit and coefficient directions. When fitting OLS on the full dataset (not cross-validated), the in-sample R^2 is still small ($R^2 \approx 0.015$), consistent with the SI results. The largest coefficient directions are typically `f0_final` (positive) and `hnr_dB` (negative), while most other features have smaller effects. Because the dataset is large (tens of thousands of frames), many p-values can become "significant" even when effect sizes are small; for this reason we emphasize predictive metrics and effect sizes rather than relying on hypothesis tests alone.

Residual diagnostics also show strong departures from normality (heavy tails), supporting the choice of MAE as a primary comparison metric.

3.4 Nonlinear shape exploration: generalized additive model (GAM)

To explore weak but potentially consistent nonlinear trends, we fit a generalized additive model (GAM). In a GAM, each predictor contributes through its own smooth function:

$$y_i = \beta_0 + \sum_{j=1}^{p} f_j(x_{ij}) + \varepsilon_i, \tag{4}$$

where $f_j(\cdot)$ is a spline-based smooth estimated from the data. Smoothness is controlled by a penalty on the spline coefficients, which discourages overly wiggly curves.

SI performance vs. descriptive value. On the full dataset, the GAM attains a modest in-sample fit (pseudo-$R^2 \approx 0.071$; MAE ≈ 1.221 Hz), suggesting that some nonlinear structure can be captured. However, under speaker-grouped CV the GAM does *not* generalize well: SI performance remains close to baseline (approximately $R^2_{\mathrm{SI}} \approx -0.04$, $\mathrm{MAE}_{\mathrm{SI}} \approx 1.24$ Hz), indicating mild overfitting rather than a transferable mapping.

Even so, the GAM is useful as a *descriptive* tool because it provides interpretable partial-effect curves. In our results:

- **HNR** shows the clearest and most stable trend: higher `hnr_dB` is associated with lower Δ_1 (a roughly monotone decrease). Across the 10–90% range of `hnr_dB`, the fitted effect size is about -3.4 Hz.
- f_0 shows weak structure overall; curvature is most apparent near the extremes where uncertainty is large and data are sparse.
- **H1–H2** contributes little in this setup, with a near-zero effect size across the central range.

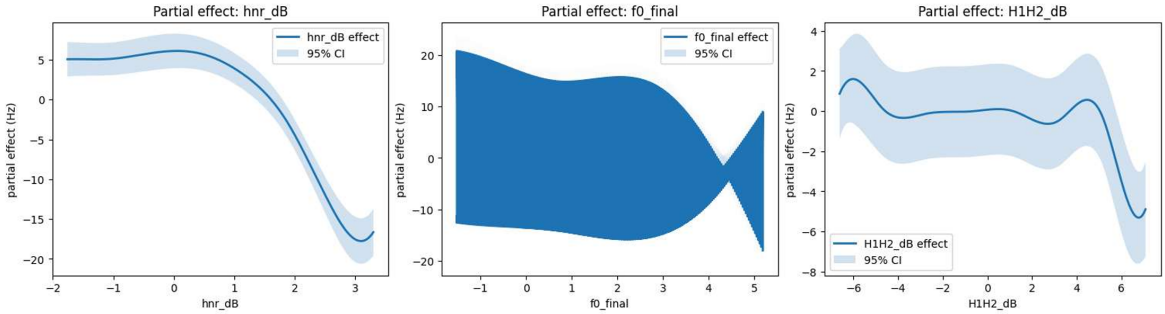


Figure 2: GAM partial effects under the speaker-independent setup. The curves are helpful for describing weak, stable trends (especially HNR), even though SI predictive performance remains limited.

3.5 Tree ensembles: nonlinear interactions as a robustness check

As a more flexible nonlinear alternative, we tested tree-based ensembles under the same speaker-grouped CV protocol. A Random Forest achieved better mean MAE in this run (mean MAE ≈ 1.05 Hz; mean $R^2 \approx 0.086$), but with noticeable variability across folds, suggesting that the gain is not consistently transferable across unseen speakers. A histogram-based gradient boosting model (HistGBDT) was closer to baseline (mean MAE ≈ 1.23 Hz; mean $R^2 \approx 0.005$). Overall, these experiments support the same conclusion as OLS/GAM: **there is no consistently strong SI predictive signal** in the tested feature set, even when allowing nonlinear interactions.

3.6 Mixed-effects modeling: separating within- and between-speaker variation

Finally, we fit linear mixed-effects models (LMMs) with a random intercept per speaker:

$$\Delta_{1,i} = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + b_{s(i)} + \varepsilon_i, \tag{5}$$

where $b_{s(i)}$ is a speaker-specific intercept for the speaker $s(i)$ of frame i. This model is useful because it explicitly separates *within-speaker* frame-to-frame variability from *between-speaker* baseline differences.

Variance decomposition. From the fitted LMM, the intraclass correlation is small but non-zero (ICC ≈ 0.0064), indicating that only a small fraction of the total variance in Δ_1 is attributable to stable between-speaker baseline shifts. Consistently, the marginal and conditional R^2 values are small (marginal $R^2 \approx 0.0095$, conditional $R^2 \approx 0.0158$), meaning that fixed acoustic covariates explain little on their own, and even after adding a random speaker intercept, most variability remains within-speaker.

Why mixed-effects does not "solve" SI prediction. Mixed-effects models can improve prediction when test data come from speakers already observed during training (or when a small amount of test-speaker data is available to estimate random effects). In our SI protocol, test speakers are *unseen*, so their random intercepts cannot be estimated and prediction necessarily relies on fixed effects only. Accordingly, under speaker-grouped CV the fixed-effects-only SI prediction remains close to baseline:

$$R^2_{\mathrm{SI}} \approx 0.00056, \qquad \mathrm{MAE}_{\mathrm{SI}} \approx 1.241 \text{ Hz}.$$

Nonlinear mixed-effects (low-degree splines). We also tested a low-degree nonlinear mixed-effects variant by adding spline bases for a few key predictors while keeping a random intercept. This increases fit slightly (marginal $R^2 \approx 0.0162$, conditional $R^2 \approx 0.0207$), but does not change the main message: **the dominant source of variation in Δ_1 is within-speaker fluctuation rather than a stable, transferable mapping from standard acoustic features.**

Table 2: Speaker-independent (SI) performance summary for predicting Δ_1 from acoustic features (20 ms frames; speaker-grouped 5-fold CV).

Model (SI evaluation)	R^2 (mean)	MAE (Hz, mean)
Median baseline (constant predictor)	-0.0127	1.249
OLS (Eq. 3)	0.0034	1.239
GAM (Eq. 4)	≈ -0.04	≈ 1.24
Random Forest	0.0857	1.048
HistGBDT	0.0050	1.232
LMM fixed-effects only (Eq. 5; SI CV)	0.00056	1.241

3.7 Takeaway

Across several model classes (linear, additive nonlinear, tree ensembles, and mixed-effects), speaker-grouped evaluation consistently shows that **standard acoustic features have limited speaker-independent explanatory power for Δ_1.** At the same time, descriptive analyses (especially GAM partial effects) suggest weak but interpretable trends—most notably a monotone relationship between HNR and Δ_1.

These findings motivate the second research question in the next section: if offsets are only weakly explained by standard measures, do speakers nevertheless exhibit distinct offset *distributions* beyond mean shifts, for example in dispersion and covariance structure?

4 Speaker dependence of harmonic-offset distributions

This section addresses the second project question: whether the harmonic-offset vector $\Delta \in \mathbb{R}^6$ follows the *same* distribution for different speakers, or whether speakers exhibit systematic differences. Importantly, we are not only interested in mean shifts. A recurring theme in this project is that offsets often differ more in *dispersion* and *covariance structure* than in simple averages.

4.1 Data used in this section

All analyses below are performed on frame-level offsets extracted from voiced, vowel-like speech frames (one frame is 20 ms). We work with the six-dimensional vector $\Delta = [\Delta_1, \ldots, \Delta_6]^\top$ (Hz), and we require a minimum

number of frames per speaker before estimating speaker-level statistics. When visualizing or running frame-level tests, we additionally cap each speaker to a maximum number of frames (sampled without replacement) to prevent a few very long recordings from dominating the results.

4.2 A simple speaker summary: mean, covariance, and total variance

For each speaker s, we compute:

- the speaker mean vector $\mu_s = \mathbb{E}[\Delta \mid s]$ (estimated by the sample mean);
- the speaker covariance $\Sigma_s = \mathrm{Cov}(\Delta \mid s)$ (estimated by the sample covariance);
- the **total variance** $\mathrm{tr}(\Sigma_s)$ as a one-number summary of dispersion.

In our filtered dataset (after balancing), we obtain **38187 frames from 138 speakers**. The distribution of total variance is heavy-tailed: the median total variance is **448.06**, the 90th percentile is **757.96**, and the maximum is **1498.50**. We will use the 90th percentile as a simple rule to define a "high-variance" group.

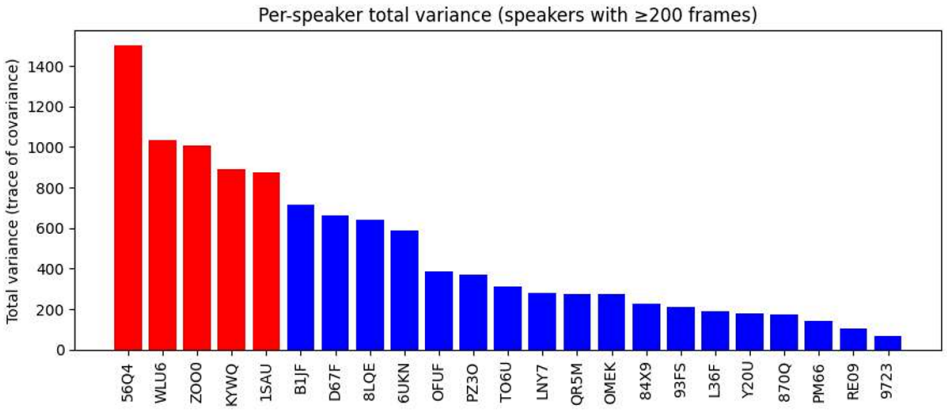


Figure 3: Histogram (or density plot) of per-speaker total variance $\mathrm{tr}(\Sigma_s)$. The vertical line marks the 90th percentile threshold used to define the high-variance group.

4.3 High-variance vs normal speakers: what differs?

Group definition. Let $\mathcal{H}$ denote the speakers with $\mathrm{tr}(\Sigma_s)$ above the 90th percentile, and $\mathcal{N}$ the remaining speakers. This split is *not* intended as a final model of speaker identity; rather, it is a compact way to test whether large dispersion speakers differ in broader distributional structure.

First look: PCA visualization (descriptive). We project the frame-level 6D offsets to 2D using PCA for visualization and color points by whether the speaker belongs to $\mathcal{H}$ or $\mathcal{N}$. This plot is descriptive only (PCA is fit on the same pooled data), but it helps check whether high-variance speakers look more spread out in a low-dimensional view.

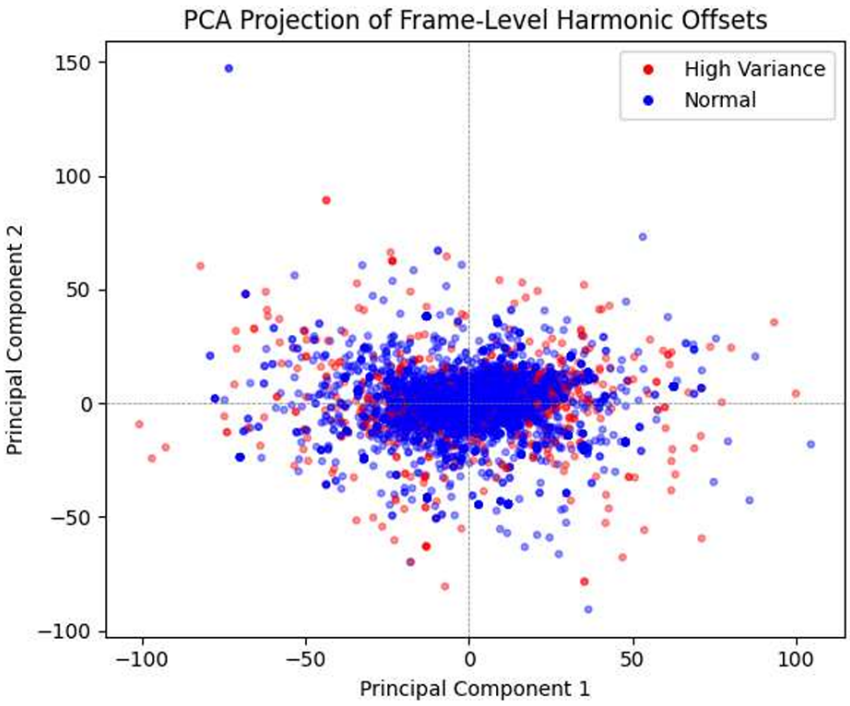


Figure 4: PCA projection of frame-level offsets (2D), colored by whether the speaker is in the high-variance group. The main visual pattern is increased spread for high-variance speakers rather than a clean mean separation.

A motivating visualization: speakers differ more in *shape* than in mean. Before introducing the formal cross-fit tests, we first provide an intuitive picture of what the between-speaker differences look like. We select the six speakers with the largest number of usable frames (to reduce sampling noise), compute the 6D offset vectors for their frames, and project all frames onto the first two principal components (PCA). For each speaker, we plot the speaker mean (dot) and a 90% covariance ellipse in the PCA plane.

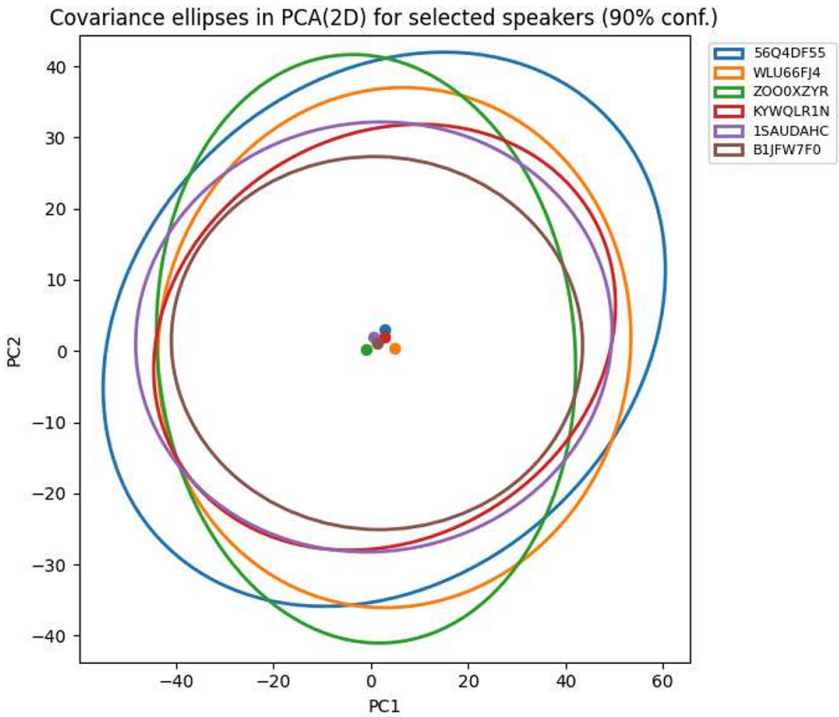


Figure 5: PCA visualization (first two PCs) for the six speakers with the most frames. Dots indicate speaker means, and ellipses show 90% covariance contours in the PCA plane. Speaker means cluster close to the origin, suggesting limited discriminative power in the mean vector, while the ellipse sizes/orientations differ visibly across speakers, indicating that the dominant between-speaker differences lie in dispersion and covariance structure (shape), rather than a simple global shift.

This figure is descriptive and should not be over-interpreted as a statistical test. In particular, if we choose groups and evaluate on the same data, we risk optimistic p-values due to "double dipping". We therefore turn to a cross-fit evaluation protocol in the next section, where group definitions are learned on one set of speakers and all hypothesis tests are carried out on held-out speakers.

4.4 Avoiding "double dipping": cross-fit evaluation

A key concern in speaker-level testing is **data leakage**: if groups are defined using the full dataset and then tested on the same data, p-values can be overly optimistic ("double dipping"). To reduce this risk, we use a **cross-fit** strategy:

1. Randomly split speakers into two disjoint sets A and B (speaker-level split).
2. Use only A to estimate the high-variance threshold (the $q = 0.9$ quantile of $\mathrm{tr}(\Sigma_s)$).
3. Apply this threshold to speakers in B to define high/normal labels in B.
4. Run statistical tests *only on* B (held-out speakers).
5. Repeat in the reverse direction ($B \to A$), and repeat across multiple random splits (seeds).

This ensures that group membership is determined out-of-sample, and the tests evaluate whether the "high-variance vs. normal" structure transfers to unseen speakers.

What each test is intended to detect. Because differences between multivariate distributions can appear in different ways, we use a small set of complementary tests:

- **Mean shift (location):** Hotelling's T^2 tests whether the two groups have different mean vectors.
- **Covariance structure (shape):** Box's M tests whether the covariance matrices differ (scale and/or correlations).
- **Dispersion (spread):** a PERMDISP-like test checks whether one group is more dispersed around its centroid.
- **Speaker-mean geometry relative to a reference:** Mahalanobis distances summarize how far each speaker mean is from a training-side reference distribution.

Interpreting results through this "map" helps separate mean-dominated differences from shape/dispersion-dominated differences.

4.4.1 Cross-fit split diagnostics and mean-shift check (Hotelling's T^2)

We repeat cross-fit over 10 random speaker splits (seeds). For each split and direction, we record (i) the learned threshold, (ii) the resulting group balance on the held-out side, and (iii) a mean-shift diagnostic using **Hotelling's** T^2 on held-out speakers.

Test idea. Hotelling's T^2 is the multivariate analogue of a two-sample t-test: it asks whether the mean offset vector differs between the two groups in $\mathbb{R}^6$. In our context, it is a *sanity check* for mean differences, not the main target, because earlier exploratory plots suggested that mean shifts are relatively small compared with variability differences.

Table 3 summarizes these diagnostics.

Table 3: Cross-fit split diagnostics and mean-shift test summary across 10 random speaker splits. Group labels on the test side are induced by a threshold learned on the training side.

	A→B (test on B)	B→A (test on A)
Hotelling T^2 significant splits ($p < 0.05$)	10/10	7/10
Fisher combined p (Hotelling T^2)	9.76×10^{-24}	1.70×10^{-11}
Prop. high on test side (median [min, max])	0.096 [0.029, 0.279]	0.107 [0.029, 0.243]
Learned threshold (median [min, max])	768.0 [600.2, 878.7]	750.3 [621.6, 879.7]

Overall, the induced high-variance group is not extremely imbalanced (median proportion around 10%), and the learned thresholds vary within a reasonable range across random splits. Hotelling's T^2 indicates that mean differences can be detectable, but the stability is asymmetric: A→B is consistently significant, while B→A shows occasional non-significant splits. This pattern already hints that *mean shift is not the most stable signature*, motivating a stronger emphasis on shape/dispersion tests below.

4.4.2 Cross-fit shape tests: covariance (Box's M) and dispersion (PERMDISP-like)

Within each cross-fit split (and for both directions A→B and B→A), we run two tests that target distributional *shape*:

Box's M (covariance equality). Box's M tests the null hypothesis that the two groups share the same covariance matrix in $\mathbb{R}^6$. In practice it is sensitive to both overall scale changes and changes in correlation structure. Because our sample size is large, extremely small p-values should be interpreted as "strong evidence of a shape difference", rather than as a measure of effect size.

PERMDISP-like dispersion test (spread around centroid). To test dispersion more directly, we compute each frame's distance to its group centroid in 6D offset space, and compare the *median* distance between groups (High vs. Normal). We assess significance using a permutation test, which avoids relying on Gaussian assumptions and provides a direct test of "one group is more spread out than the other". Alongside the p-value, we also report the dispersion *effect sign* (whether High−Normal is positive).

Table 4 summarizes the key results across random splits.

Table 4: Cross-fit shape/dispersion test summaries across 10 random splits.

	A→B	B→A
Box's M median p	≈ 0	≈ 0
PERMDISP median p	3.33×10^{-4}	3.33×10^{-4}
Dispersion effect size sign (High−Normal)	positive in all splits	positive in all splits

These cross-fit results strongly support that group differences are dominated by *covariance/shape* and *dispersion* rather than a pure mean shift: Box's M remains extreme across splits (often reaching numerical underflow), and PERMDISP-like p-values repeatedly hit the permutation-resolution floor, while the dispersion effect is consistently positive.

4.4.3 Cross-fit Mahalanobis tests on speaker means

A complementary speaker-level view is to ask whether test speakers' *mean offset vectors* sit differently relative to a reference distribution learned on the training side. For each test speaker, we compute a Mahalanobis distance (MD) from its mean vector to the training-side reference distribution. If high-variance speakers also have "atypical" mean locations relative to the training distribution, their MDs should tend to be larger.

We compare MD distributions between groups on the held-out side using (i) a Mann–Whitney test and (ii) a centroid-based permutation test. Table 5 summarizes the results.

Table 5: Cross-fit Mahalanobis-distance (MD) tests on speaker means across 10 random splits.

	A→B	B→A
Mann–Whitney on MD (median p; frac($p < 0.05$))	4.93×10^{-3}; 1.00	8.67×10^{-5}; 0.90
Centroid permutation (median p; frac($p < 0.05$))	1.40×10^{-2}; 0.80	4.80×10^{-3}; 0.90

These MD-based tests indicate that mean-vector geometry differs to some extent as well, but the most stable and strongest evidence still comes from dispersion/covariance differences (Table 4).

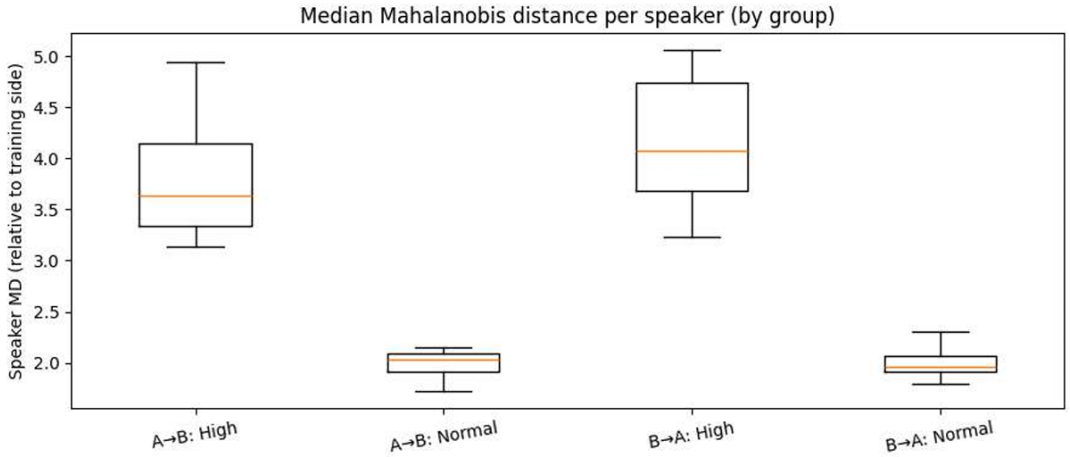
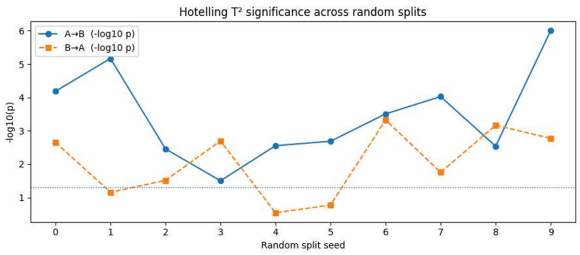


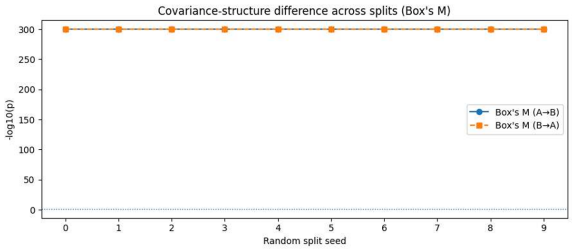
Figure 6: Cross-fit comparison of per-speaker Mahalanobis distances (MD) by group. Each box summarizes the distribution of speaker-level MDs under A→B and B→A evaluation.

4.4.4 Visual stability across random splits

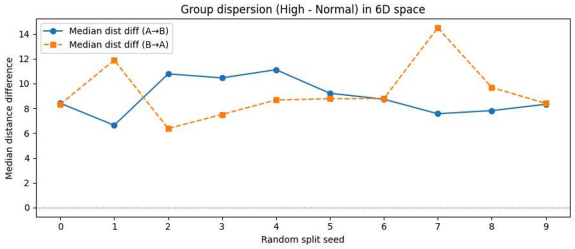
To complement the summary tables, Figure 7 visualizes how evidence varies across the 10 random splits. The key takeaway is that mean-shift evidence (Hotelling's T^2) is present but more split-dependent, whereas the shape-based evidence (Box's M and PERMDISP-like dispersion) is consistently strong, both in significance and in the direction of the dispersion effect.



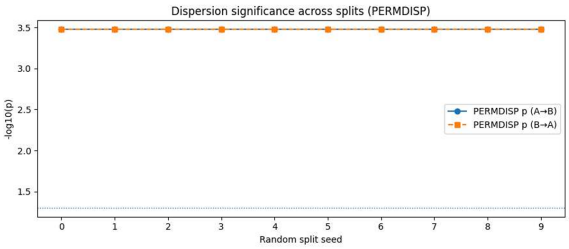
(a) Hotelling T^2 across splits ($-\log_{10} p$), testing mean shift.



(b) Box's M across splits ($-\log_{10} p$), testing covariance structure.



(c) Dispersion effect size (High−Normal): median distance-to-centroid difference.



(d) PERMDISP-like permutation p-values across splits ($-\log_{10} p$).

Figure 7: Cross-fit stability across 10 random speaker splits. The dotted line marks $p = 0.05$. Box's M p-values often underflow to numerical zero, while PERMDISP p-values frequently hit the permutation-resolution floor (here $p_{\min} \approx 3.33 \times 10^{-4}$), indicating consistently strong evidence for dispersion/covariance differences.

4.5 Take-away

Across multiple complementary analyses, we find strong evidence that offset distributions are speaker-dependent. The most stable and visually clear differences are **not** simple mean shifts, but rather differences in **dispersion** and **covariance structure**: some speakers have much more variable offsets and different cross-harmonic dependency patterns. This motivates treating speaker identity as a key factor in later longitudinal or clinical analyses, and it also explains why purely speaker-independent prediction of offsets from standard acoustic features is challenging in practice.

5 Summary and outlook

This report studied small deviations from an ideal harmonic spectrum in voiced speech, quantified as $\mathbf{\Delta} = [\Delta_1, \ldots, \Delta_6]^\top$ with $\Delta_n = f_n - nf_0$. Using a frame-level pipeline (20 ms voiced, vowel-like frames), we asked two questions: (i) whether harmonic offsets can be explained by standard acoustic measures in a *speaker-independent* way, and (ii) whether different speakers exhibit genuinely different offset *distributions* beyond simple mean shifts.

First, Speaker-independent feature–offset relationships are weak. Across several model classes (OLS, GAM, tree ensembles, and mixed-effects), speaker-grouped cross-validation consistently showed limited out-of-speaker predictive performance for Δ_1. Linear regression was only marginally better than a "no-feature" constant baseline, and more flexible models did not yield a stable improvement that transferred across unseen speakers. At the same time, the nonlinear GAM analysis was still useful descriptively: it suggested weak but interpretable trends, with HNR showing the clearest monotone association with Δ_1 in the pooled data, even though this structure did not translate into strong speaker-independent prediction.

Second, Offset distributions differ across speakers, mainly in dispersion and covariance (shape). In contrast to the weak feature-based explainability, speaker-level analyses provided strong evidence that offset distributions vary systematically across individuals. A PCA visualization of the highest-frame speakers suggested that speaker means cluster relatively closely, whereas covariance ellipses differ substantially in size and orientation, motivating a focus on distributional *shape* rather than mean. Formal tests reinforced this: under a cross-fit evaluation protocol designed to avoid "double dipping", covariance-structure tests (Box's M) and dispersion tests (PERMDISP-like) remained consistently significant across random speaker splits, indicating robust differences in variability and correlation structure in the 6D offset space. Mean-shift evidence (e.g., Hotelling's T^2) was present but less stable across splits, supporting the interpretation that speaker dependence is expressed more strongly through dispersion/covariance than through a global mean offset.

Taken together, the findings point to a clear separation: standard acoustic features capture at most a small, weakly transferable component of offset variability at the frame level, while speaker identity leaves a strong, multivariate signature on the *distributional shape* of offsets. This also helps reconcile the mixed-effects results: a random intercept can capture only mean-like baseline shifts, whereas the dominant between-speaker differences observed here appear to lie in covariance/dispersion patterns.

Limitations. Several limitations are worth noting. First, the speaker-independent modeling focused on Δ_1 as the most stable offset component; extending the same analysis to the full vector $\mathbf{\Delta}$ is an important next step. Second, frames were extracted as vowel-like voiced segments but vowel identity was not explicitly separated, which may blur structure if different vowels have systematically different offset behavior. Third, our mean-shift check relied on Hotelling's T^2, which in its standard two-sample form assumes equal covariance matrices across groups (via a pooled covariance estimate). In our setting, however, the covariance structure is precisely what appears to differ strongly between groups (as indicated by Box's M), so this assumption is unlikely to hold. As a consequence, the Hotelling T^2 p-values should be interpreted cautiously: they can be anti-conservative (too optimistic) when covariances are unequal, and therefore should not be treated as primary evidence compared to the shape/dispersion results.

Future work. There are three natural directions forward. (i) Move from a single component to genuinely multivariate modeling of $\mathbf{\Delta}$, for example using PCA/PLS projections, multivariate regression, or distributional embeddings. (ii) Replace the equal-covariance mean test with alternatives that remain valid under unequal covariance, such as heteroscedastic multivariate two-sample tests (Welch/James-type corrections) or a permutation-based test on mean differences that does not rely on pooled-covariance assumptions. (iii) Apply the same pipeline to clinical longitudinal data to study within-patient change over longer time scales. For example, for patients with known intervention dates (e.g., surgery), one can track offset trajectories pre/post intervention and test whether the offset distribution (mean and/or dispersion/covariance) shows a systematic shift around the event date, while controlling for speaker-specific baselines.

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